Problem 1 | Problems T5_MechanicalProperties.pdf #1 | Tensile test · full stress-strain analysis

Complete Analysis of a Tensile Test from the Load-Elongation Table

Original statement · Problems T5_MechanicalProperties.pdf

1. A cylindrical aluminum sample has a diameter of 12.8 mm and an initial base length of 12.5 mm and it is subjected to uniaxial tension. Use the data from the load/elongation table to answer the following questions:

- *a. Represent the engineering stress data versus the engineering strain.*
- *b. Calculate Young's modulus.*
- *c. Calculate the yield strength, assuming that the residual strain at the yield point is 0.002.*
- *d. Work out the tensile strength of this alloy.*
- *e. Determine the approximate value of ductility expressed as a deformation percentage.*
- *f. Determine the approximate value of the modulus of resilience.*
- *g. Calculate the load (N) and the elongation (mm) of the sample at the yield point.*
- *h. Calculate an approximate working stress for this material.*
- *i. With the values obtained, is it possible to calculate the load applied to the sample if the elongation produced is 0.1 mm? If it is possible, work it out and if it is not possible, explain your answer and provide an alternative method.*

Point	F (N/100)	ΔL (mm)
1	127	0.006
2	254	0.012
3	381	0.019
4	508	0.025
5	762	0.038
6	891	0.051
7	927	0.063
8	1025	0.095
9	1078	0.127
10	1194	0.190
11	1283	0.254
12	1497	0.508
13	1590	0.762
14	1604	0.889
15	1595	1.016
16	1515	1.270

Point	F (N/100)	ΔL (mm)
17	1247	1.460

(The sheet prints the table in two rows, F and ΔL, followed by the word "Fracture"; here each point has its own row. Point 17 is the last record before the fracture.)

Official Solution: b. $E \approx 200$ **GPa; c.** $\sigma_y = 740$ **MPa; d.** $\sigma_{max} = 1250$ **MPa; e. 11.3 %; f.** $U_r = 1.37$ **MPa; g.** $F = 9.5 \times 10^4$ **N,** $\Delta l = 0.071$ **mm; h.** $\sigma_w = 370$ **MPa; i. it can only be obtained graphically,** $F = 10.3 \times 10^4$ **N**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** The load acts along the axis of the cylinder, and up to the necking the stress is the same on every cross-section of the gauge length. Under this hypothesis the engineering stress and strain are defined with the original area and length [Session 11 Slide 9].
2. **Linear elastic start.** For small loads the curve is a straight line through the origin, Hooke's law $\sigma = E\varepsilon$, and the slope of that line is the Young's modulus [Session 11 Slide 13].
3. **Conventional yield strength.** The yield strength is the stress at which the curve crosses the line drawn parallel to the elastic line from the residual strain 0.002 (0.2 % offset), as the statement asks [Session 11 Slide 28].
4. **Between table rows the curve is a straight segment.** The table gives discrete points, so every reading between two of them is a linear interpolation.
5. **Elastic recovery.** When the sample breaks the load disappears and the elastic strain is recovered, so the ductility is the plastic strain at fracture [Session 11 Slide 32].
6. **Safety factor.** The statement gives none; the slide says that N is usually 2, and that value is used for the working stress [Session 11 Slide 29].

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	12.8 mm	12.8×10^{-3} m
Initial gauge length	l_0	12.5 mm	12.5×10^{-3} m
Load (table)	F	$100 \times (F/100)$	N
Elongation (table)	ΔL	see table	mm
Safety factor	N	2	

Unknowns: the curve (a), E (b), σ_y (c), σ_{max} (d), the ductility (e), U_r (f), F_y and Δl_y (g), σ_w (h), and the load at $\Delta L = 0.1$ mm (i).

PHASE 2 · FRAMES & STRATEGY

There is a single direction, the axis of the sample, and the stress state is uniaxial, so no change of basis and no rotation matrix are involved.

Strategy. Every item comes from the same table of engineering stress and strain, so the first thing to do is to build it:

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

- The Young's modulus is the slope of the straight initial part (Hooke's law). A straight line through the origin is fitted by least squares: it minimises $\sum(\sigma_i - E\varepsilon_i)^2$, which gives $E = \sum \sigma_i\varepsilon_i / \sum \varepsilon_i^2$.
- The yield strength is found by the 0.2 % offset construction, solved algebraically on the segment of the table where the curve crosses the offset line.
- The tensile strength is the maximum nominal stress, and since A_0 is constant it corresponds to the maximum load.
- The ductility is the plastic strain at fracture, and the modulus of resilience is the area under the elastic line up to the yield point, $U_r = \sigma_y^2/(2E)$ [Session 11 Slide 35].
- For the load at a given elongation the table is used again: Hooke's law only holds in the elastic zone.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(12.8\ \text{mm})^2}{4}$$

The square is $12.8^2 = 163.84$, then $\pi \times 163.84 = 514.72$ and dividing by 4:

$$A_0 = 128.68\ \text{mm}^2 = 1.2868 \times 10^{-4}\ \text{m}^2$$

2. Engineering stress and strain of every point

The load of the table is in hundreds of newtons, so $F = 100 \times (F/100)$. For point 1:

$$F_1 = 100 \times 127 = 12\,700\ \text{N} \qquad \sigma_{n,1} = \frac{12\,700\ \text{N}}{128.68\ \text{mm}^2} = 98.69\ \text{MPa} \qquad \varepsilon_{n,1} = \frac{0.006\ \text{mm}}{12.5\ \text{mm}} = 4.80 \times 10^{-4}$$

The same operation for all the points (the last column is the secant modulus, used in item b):

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	12 700	0.006	98.69	0.00048	205.6
2	25 400	0.012	197.39	0.00096	205.6
3	38 100	0.019	296.08	0.00152	194.8
4	50 800	0.025	394.78	0.00200	197.4
5	76 200	0.038	592.17	0.00304	194.8
6	89 100	0.051	692.42	0.00408	169.7
7	92 700	0.063	720.39	0.00504	142.9
8	102 500	0.095	796.55	0.00760	104.8
9	107 800	0.127	837.74	0.01016	82.5
10	119 400	0.190	927.89	0.01520	61.0
11	128 300	0.254	997.05	0.02032	49.1
12	149 700	0.508	1163.35	0.04064	28.6
13	159 000	0.762	1235.63	0.06096	20.3
14	160 400	0.889	1246.51	0.07112	17.5
15	159 500	1.016	1239.51	0.08128	15.2
16	151 500	1.270	1177.34	0.10160	11.6
17	124 700	1.460	969.07	0.11680	8.3

a) Stress-strain curve

The points of the table are plotted as they are, joined by straight segments. The right-hand panel enlarges the elastic part, where the Young's modulus line and the offset line of item c are drawn.

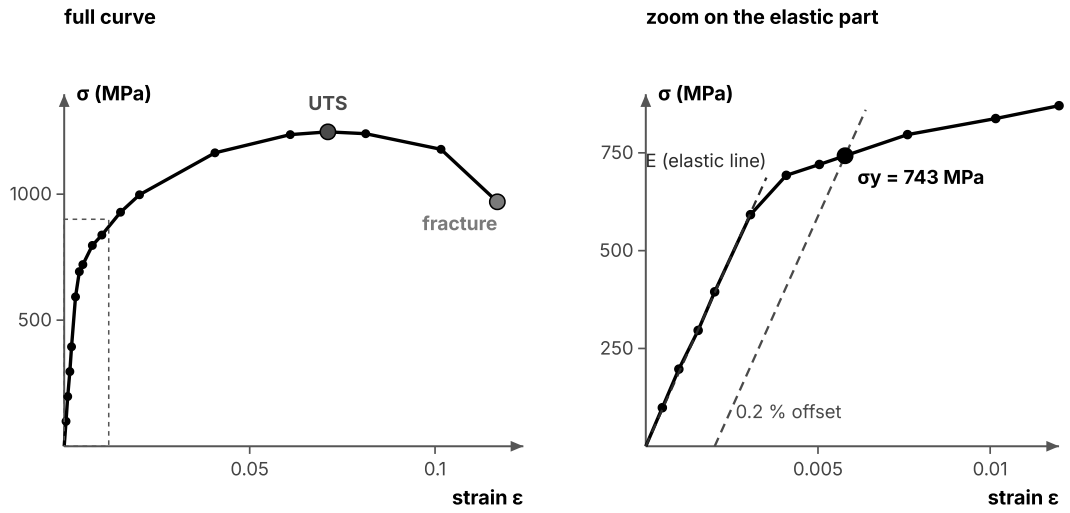


Figure 1. Problem 1: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

b) Young's modulus

The secant modulus σ_n/ε_n of the last column is almost constant, 195 to 206 GPa, for points 1 to 5, and it drops to 170 GPa at point 6 and to 143 GPa at point 7: the specimen leaves the linear zone after point 5. The straight part is therefore made of points 1 to 5, and the fit through the origin is:

Point	ε_i	σ_i (MPa)	$\sigma_i\varepsilon_i$ (MPa)	ε_i^2
1	0.00048	98.69	0.0474	2.304×10^{-7}
2	0.00096	197.39	0.1895	9.216×10^{-7}
3	0.00152	296.08	0.4500	2.310×10^{-6}
4	0.00200	394.78	0.7896	4.000×10^{-6}
5	0.00304	592.17	1.8002	9.242×10^{-6}

$$\sum \sigma_i\varepsilon_i = 3.277 \text{ MPa} \qquad \sum \varepsilon_i^2 = 1.6704 \times 10^{-5}$$

$$E = \frac{\sum \sigma_i\varepsilon_i}{\sum \varepsilon_i^2} = \frac{3.277}{1.6704 \times 10^{-5}} = 196160 \text{ MPa} = \mathbf{196.2\ GPa} \approx 2.0 \times 10^{2} \text{ GPa}$$

c) Yield strength (0.2 % offset)

The offset line starts at $\varepsilon = 0.002$ and is parallel to the elastic line:

$$\sigma_{off}(\varepsilon) = E\,(\varepsilon - 0.002)$$

The curve is above the line while $g = \sigma_n - E(\varepsilon_n - 0.002)$ is positive. The sign of g changes between points 7 and 8, so the crossing is on that segment:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
6	0.00408	692.4	408.0	284.4
7	0.00504	720.4	596.3	124.1
8	0.00760	796.6	1098.5	-301.9
9	0.01016	837.7	1600.7	-762.9

The segment between the two points is the line

$$\sigma(\varepsilon) = \sigma_7 + m\,(\varepsilon - \varepsilon_7) \qquad m = \frac{\sigma_8 - \sigma_7}{\varepsilon_8 - \varepsilon_7} = \frac{796.55 - 720.39}{0.00760 - 0.00504} = 29749.3\ \text{MPa}$$

Equating it to the offset line, $\sigma_7 + m(\varepsilon - \varepsilon_7) = E(\varepsilon - 0.002)$, and solving for ε:

$$\varepsilon^{*} = \frac{\sigma_7 - m\,\varepsilon_7 + 0.002\,E}{E - m} = \frac{720.39 - 29749.3 \times 0.00504 + 0.002 \times 196160}{196160 - 29749.3} = \frac{962.778}{166411.2} = 0.0057855$$

$$\sigma_y = E\,(\varepsilon^{*} - 0.002) = 196160 \times (0.0057855 - 0.002) = \mathbf{742.6\ MPa}$$

d) Tensile strength

The maximum load of the table is $F_{max} = 160\,400$ N at point 14. The nominal stress uses the original area, so the maximum load gives the maximum stress:

$$\sigma_{max} = \frac{F_{max}}{A_0} = \frac{160\,400\ \text{N}}{128.68\ \text{mm}^2} = \mathbf{1246.5\ MPa}$$

e) Ductility

The last record before the fracture is point 17: $\Delta L = 1.460$ mm and $F = 124\,700$ N, so

$$\varepsilon_f = \frac{1.460}{12.5} = 0.1168 \qquad \sigma_f = \frac{124\,700}{128.68} = 969.1 \text{ MPa}$$

At that moment the strain is the sum of an elastic and a plastic part, $\varepsilon_t = \varepsilon_e + \varepsilon_p$. After the fracture the load is zero and the elastic part is recovered along a line parallel to the elastic one, so what remains is ε_p [Session 11 Slide 32]:

$$\varepsilon_e = \frac{\sigma_f}{E} = \frac{969.1}{196160} = 0.00494 \qquad \varepsilon_p = \varepsilon_f - \varepsilon_e = 0.1168 - 0.00494 = 0.1119$$

$$EL\,(\%) = \varepsilon_p \times 100 = \mathbf{11.2\ \%}$$

f) Modulus of resilience

The area under the elastic line up to the yield point is a triangle, $U_r = \tfrac{1}{2}\sigma_y \varepsilon_y$ with $\varepsilon_y = \sigma_y / E$ [Session 11 Slide 35]:

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(742.6 \text{ MPa})^2}{2 \times 196160 \text{ MPa}} = \mathbf{1.41\ MPa}$$

($1 \text{ MPa} = 1 \text{ MJ/m}^3$.)

g) Load and elongation at the yield point

The load is the yield stress times the original area, and the elongation is the total strain of the yield point (the value ε^* of item c) times the original length:

$$F_y = \sigma_y A_0 = 742.6 \text{ MPa} \times 128.68 \text{ mm}^2 = 95\,554 \text{ N} = \mathbf{95.6\ kN}$$

$$\Delta l_y = \varepsilon^* l_0 = 0.0057855 \times 12.5 \text{ mm} = \mathbf{0.0723\ mm}$$

h) Working stress

$$\sigma_w = \frac{\sigma_y}{N} = \frac{742.6}{2} = \mathbf{371.3\ MPa}$$

i) Load for an elongation of 0.1 mm

The elongation of the yield point is 0.0723 mm, so 0.1 mm is in the plastic zone and the load is not proportional to the elongation. Using Hooke's law anyway, $F = EA_0\Delta l/l_0$, would give

$$F = \frac{196160\ \mathrm{N/mm^2} \times 128.68\ \mathrm{mm^2} \times 0.1\ \mathrm{mm}}{12.5\ \mathrm{mm}} = 201\,935\ \mathrm{N} = 202\ \mathrm{kN}$$

which is larger than the maximum load that the sample supports, 160 400 N: the answer is **not** possible with the values obtained. The alternative is the table itself. The elongation 0.1 mm is between points 8 ($\Delta L = 0.095$ mm, $F = 102\,500$ N) and 9 ($\Delta L = 0.127$ mm, $F = 107\,800$ N), and a linear interpolation on that segment gives

$$F = F_8 + \frac{0.1 - 0.095}{0.127 - 0.095}\,(F_9 - F_8) = 102\,500 + 0.15625 \times (107\,800 - 102\,500) = \mathbf{103\,328\ N}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\sigma_n = F/A_0$ in N/mm² is MPa, ε_n is a ratio of lengths and has no units, so $E = \sigma/\varepsilon$ is in MPa and $U_r = \sigma_y^2/2E$ is in MPa (energy per unit volume).
- **Order.** $\sigma_y = 742.6$ MPa $< \sigma_{max} = 1246.5$ MPa, and the fracture stress (969.1 MPa) is below the maximum, as in a ductile sample that necks.
- **The fit.** The secant moduli of points 1 to 5 are within 4.8 % of the fitted E, and the offset line crosses the curve between points 7 and 8, inside the segment used.
- **Item i.** The interpolated load, 103.3 kN, is between the loads of the two rows and below the maximum load; the Hooke value, 202 kN, is not.

Interpretation

The sample is stiff ($E \approx 196$ GPa) and strong, and it yields at about 742.6 MPa, a strain of 0.579 %. From there to the maximum load the strain hardening raises the stress by 504 MPa, and the 11.2 % of plastic elongation at fracture shows that it is also ductile.

THE DATA ARE NOT THOSE OF ALUMINIUM

The table of Session 11 gives $E = 69$ GPa for aluminium and 207 GPa for steel [Session 11 Slide 17]. The data of this problem give $E \approx 196$ GPa and a yield strength above 700 MPa, which are the values of a steel, not of aluminium. The statement says "aluminum"; the data are used as they are.

Check against the official solution

The key printed on the sheet is compared with the values computed here:

Item	Official	Computed	Comment
b. E	≈ 200 GPa	196.2 GPa	agrees (2 %)
c. σ_y	740 MPa	742.6 MPa	agrees
d. σ_{max}	1250 MPa	1246.5 MPa	agrees (rounded)
e. ductility	11.3 %	11.2 %	agrees within the reading
f. U_r	1.37 MPa	1.41 MPa	agrees (2.5 %); the key uses $E = 200$ GPa and $\sigma_y = 740$ MPa
g. F_y, Δl_y	9.5×10^4 N, 0.071 mm	95 554 N, 0.0723 mm	agrees
h. σ_w	370 MPa	371.3 MPa	agrees
i. load at 0.1 mm	10.3×10^4 N, graphically	103\,328 N	agrees

READING OF THE TABLE

The text of the PDF prints the first load of the table as "1127", but the figure of the sheet shows "127". With 127 the first point gives $\sigma/\varepsilon = 206$ GPa, in line with the rest of the elastic points and with the key, so 127 (12 700 N) is used.